

Nicoló Dalmasso

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TECHNICAL SKILLS

- **Programming & Tools:** Python (Jupyter Ecosystem, NumPy, Pandas, Matplotlib, SciPy), SQL, Git/GitHub, GitHub Actions (CI/CD), Docker.
- **Data Engineering:** ETL pipelines, feature engineering, workflow automation, performance optimization.
- **Data Analysis:** Spatial statistics (point pattern analysis, spatial clustering), exploratory analysis (EDA), data visualization, anomaly detection.
- **Statistical Modeling:** Bayesian inference, MCMC sampling, Monte Carlo simulations, probabilistic modeling, hypothesis testing.
- **ML Techniques:** Supervised/unsupervised learning, neural networks, model training and fine-tuning, cross-validation.
- **AI Tools:** LLM evaluation, prompt engineering, GenAI platforms (GPT, Claude).

PROFESSIONAL EXPERIENCE

Data Scientist **May 2026 - Present**
AURIN (Australian Urban Research Infrastructure Network)

- **Data Analysis & Curation** Conduct data exploration, processing, validation, and statistical analysis across priority initiatives, producing trustworthy, FAIR-compliant data assets for research and product stakeholders.
- **Geospatial Engineering** Develop and maintain Python scripts, tools, and notebooks for spatial analysis, modelling, and visualisation within reproducible, modular, versioned analytical workflows.
- **Cross-functional Delivery** Operate within an agile, squad-based delivery model alongside product, engineering, and research teams, supporting sprint execution and data-driven initiatives.
- **Documentation** Produce technical and analytical documentation to support delivery continuity, knowledge transfer, and reuse of analytical outputs across internal teams.

LLM Training Specialist **Sep 2025 - Apr 2026**
Mercor Intelligence

- **Problem Design** Design advanced STEM reasoning problems to test frontier LLM capabilities and identify failure modes in technical reasoning.
- **Performance Analysis** Evaluate model outputs using statistical methods to quantify reasoning gaps, tracking and categorizing failure modes to inform data-driven improvements.
- **Solution Development** Develop step-by-step solutions with detailed rubrics and measurable success criteria for evaluating model responses on state-of-the-art technical problems.
- **Quality Assurance** Validate model accuracy through expert review and statistical analysis, ensuring consistency and reliability across diverse problem categories.

Doctoral Researcher **Mar 2022 - Sep 2025**
The University of Melbourne

- **Spatial Analysis & Visualization** Performed large-scale morphological classification and merger analysis on high-resolution imaging data, applying point pattern analysis and spatial clustering to produce publication-quality visualizations and translate complex findings into clear narratives for diverse audiences.
- **Data Engineering & Analysis** Developed automated Python pipelines for processing heterogeneous datasets, reducing manual processing time by 70%.
- **Machine Learning & Statistical Modeling** Built probabilistic models using Bayesian inference, MCMC sampling, and Monte Carlo methods, improving benchmark accuracy by 15% via ensemble methods and statistical validation.
- **Algorithm Optimization** Reduced computational runtime by 40% through profiling and code optimization, enabling deployment of analysis workflows.
- **Research Leadership & Collaboration** Led two international multi-institutional collaborations, delivering 7 peer-reviewed publications (5 first-author) in Nature Astronomy and leading journals.

Teaching Assistant **Mar 2022 - Sep 2025**
The University of Melbourne

- **Teaching & Mentorship** Conducted tutorials and lectures for undergraduate and graduate Physics and Astrophysics courses, translating complex concepts into accessible explanations.
- **Assessment** Evaluated assignments, projects and exams for 200+ students, tracking grade distributions to ensure grading consistency and identify opportunities for targeted support.
- **Outreach** Co-delivered Year 12 Physics lessons through the SEAMS University outreach program, adapting technical content for high school audiences.

EDUCATION

The University of Melbourne (AU)

- **Doctor of Philosophy (PhD) in Physics – Astrophysics**

Mar 2022 - Sep 2025

Università degli Studi di Torino (IT)

- **MSc in Astrophysics and Theoretical Physics** - 110/110 with distinction
- **BSc in Physics**

Oct 2019 - Oct 2021

Oct 2016 - Oct 2019